

Deep Learning Optimization Report

A Professional Record of Neural Network Tuning & Performance

Experiment Results Overview

Model	Key Architecture & Optimization	Val Acc
Model 0	Linear baseline, Adam Optimizer	92.16%
Model 1	1 Hidden Layer (128 units), ReLU	97.63%
Model 2	Dropout 0.2, EarlyStopping (Patience 3)	97.82%
Model 3	Dropout 0.2 (Multi-layer), ES (Patience 5)	97.75%
Model 4	Dropout 0.3, ES + restore_best_weights	97.57%
Model 5	ReduceLROnPlateau + EarlyStopping	97.95%

Technical Fundamentals

ReLU Activation: The function $f(x) = \max(0, x)$ is the industry standard for hidden layers. It enables the model to solve complex non-linear problems efficiently.

Dropout Regularization: By randomly deactivating neurons, the model is forced to find multiple paths to the correct answer, which significantly reduces overfitting.

Strategic Recommendations

```
# Optimal Model 5 Callback Configuration
early_stop = EarlyStopping(monitor='val_loss', patience=6,
restore_best_weights=True)
reduce_lr = ReduceLROnPlateau(monitor='val_loss',
factor=0.5, patience=2, min_lr=0.00001)
```

This configuration balances "exploration" and "exploitation"—stopping training when improvement ceases, but also sharpening the search by lowering the learning rate as the model approaches peak accuracy.